

Sales Navigator Lead Generation

Pipeline: Step-by-Step Walkthrough

This document walks through the technical execution flow of the B2B prospecting pipeline when running [orchestrator.js](#).

1. Step-by-Step Execution Flow

```
graph TD
    A[1. Parse Options & Load Context] --> B[2. Run Playwright Scraper Loop]
    B --> C[3. Extract Employee Profiles]
    C --> D[4. Convert Recording WebM to MP4]
    D --> E{5. Are Prospects Found?}
    E -- Yes --> F[6. Send Batch to Gemini API]
    E -- No --> G[Skip Gemini & Write Empty CSV]
    F --> H[7. Write CSV & Individual Drafts]
    G --> I[Write Runs Logs]
    H --> I
```

Step 1: Initialization & Context Loading

- **CLI Parameter Reading:** Parses command-line flags like target companies (e.g. `--companies "Pentagram,Order"`), scraping depth (`--limit`), visual headed mode (`--headed`), and real runs (`--real` vs `--dry-run` mock mode).
- **Context Ingestion:** Loads [company_profile.md](#) (Emergentic's service positioning around AI event simulations) and [outreach_guidelines.md](#) (standards for quiet, low-hype, short-form copywriting).

Step 2: The Scraper Adapter Loop

For each target company, the script executes the scraper in [sales_nav_adapter.js](#):

- **Browser Launch:** Launches Playwright Chromium with a stealth plugin to prevent fingerprint profiling.

- **Session Restoration:** Automatically loads authentication state cookies from [auth.json](#) to bypass standard login pages.
- **Recording Initialization:** Configures visual capture to record the run.
- **Filter Interaction:**
 1. Navigates directly to the empty Lead Search panel.
 2. Dismisses any blocking walkthrough popovers (coach-marks).
 3. Expands the **Current company** accordion filter.
 4. Clicks the text box and types the name letter-by-letter to trigger the search listeners.
 5. Performs case-insensitive matching against autocomplete suggestions to click the exact target company (e.g. selecting `COLLINS` instead of `Collins Aerospace`).
- **Human-like Scrolling:** Staggers page scrolling to dynamically load all search result list items.

Step 3: Profile Extraction

For each person on the results list, it extracts:

- **Name & Job Title:** Parsed from DOM list elements.
- **Sales Nav URL:** Parsed from relative links.
- **Mutual Connections:** Scans for shared network status text.

Step 4: Screen Recording Conversion

Once the browser context is closed, the script uses a child process to call `ffmpeg` to transcode Playwright's native `.webm` screen recording into an `.mp4` video inside the campaign folder, deleting the source webm file upon completion.

Step 5: Batch AI Evaluation

All scraped profiles are compiled into a single array:

- **Guard Check:** If the profile count is 0, the script skips the LLM API call entirely.
- **Single Batch call:** Sends the profiles, company details, and outreach guidelines to the **Gemini API** (`gemini-1.5-flash`) requesting a structured JSON payload containing a fit score, custom angles, and a cold letter draft for each person.

- **Cost Tracker:** Logs the exact session and cumulative token counts + calculated USD pricing.

Step 6: Exporting Outputs

Writes all compiled data into target campaign outputs:

- **CSV Spreadsheet:** Generates [scraped_leads.csv](#) with columns for name, title, url, connections, score, notes, and outreach letter draft.
- **Markdown Files:** Saves individual text drafts under the `drafts/` directory for manual inspection.
- **Execution Logs:** Saves `run_log.json` recording total execution time, targets, and any scraper errors.